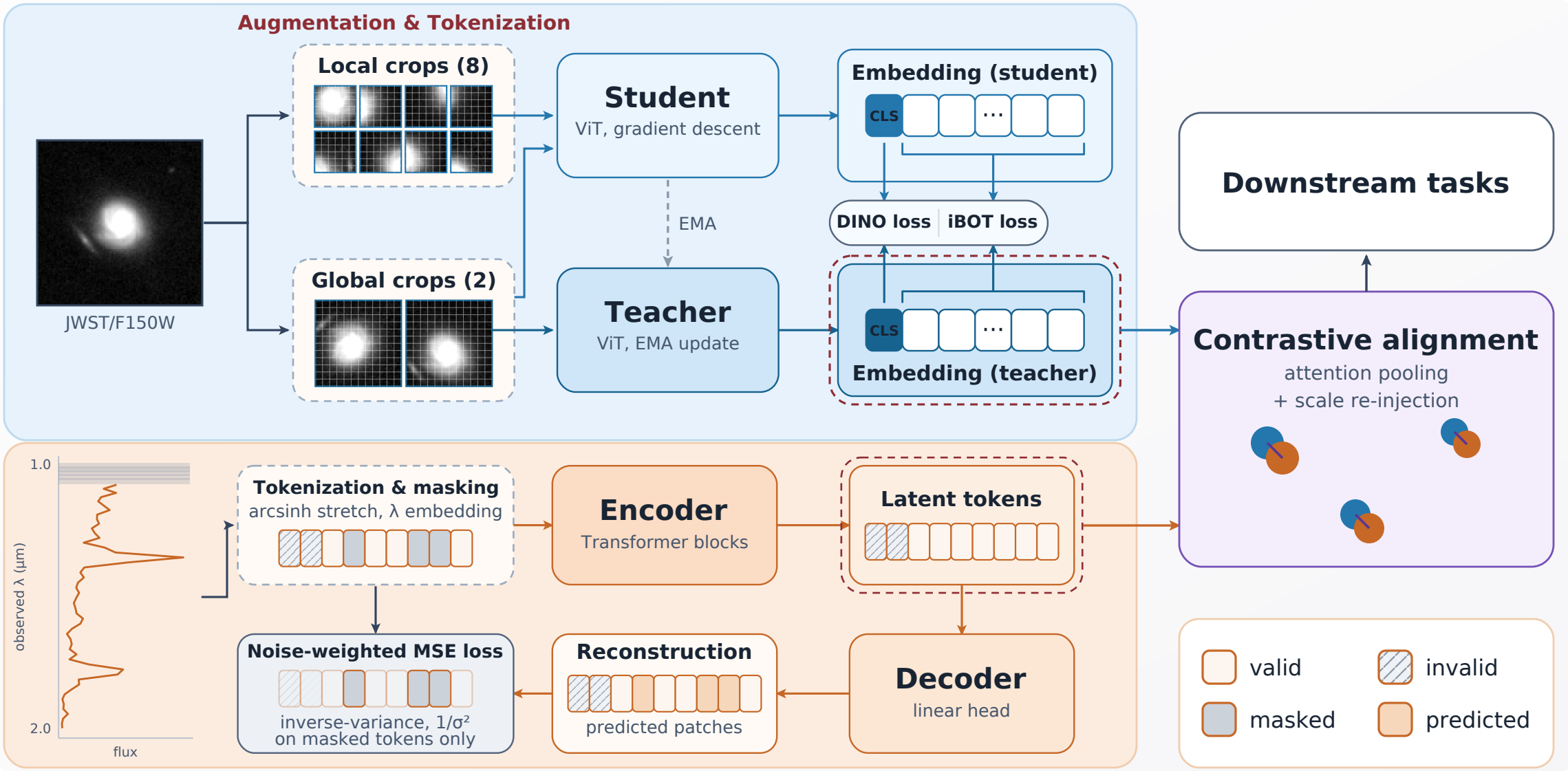

Augmentation & Tokenization
JWST/F150W
Local crops (8)
Global crops (2)
Student
ViT, gradient descent
EMA
Teacher
ViT, EMA update
Embedding (student)
CLS
Embedding (teacher)
DINO loss
iBOT loss
Downstream tasks
Contrastive alignment
attention pooling
+ scale re-injection
1.0
2.0
observed λ (μm)
flux
Tokenization & masking
arcsinh stretch, λ embedding
Encoder
Transformer blocks
Latent tokens
Noise-weighted MSE loss
inverse-variance, 1/σ²
on masked tokens only
Reconstruction
predicted patches
Decoder
linear head
valid
invalid
masked
predicted